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RESEARCH

Core-shell structure Fe_2O_3@C nanocomposite anode material prepared from the leaching solution of iron concentrate for lithium-ion batteries

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Abstract
Core-shell structure Fe_2O_3@C composite by utilizing the iron concentrate leachate with a selective chemical precipitation method and illustrate its great potential as a high-performance anode material of lithium-ion batteries (LIBs). The Fe_2O_3@C composite exhibits superior cyclic performance (723.08 mA h g^{-1} over 300 cycles at 1 A g^{-1}) and high-rate capability (322.01 mA h g^{-1} even at a high current density of 5 A g^{-1}) in a half cell. The excellent performance of the Fe_2O_3@C can be attributed to particles contain abundant internal voids, which facilitates the infiltration of the electrolyte and alleviates the volume change of electrode material, thereby enhancing the electrochemical performance of the material. Due to the unique core-shell structure nanostructure, Fe_2O_3@C composite shows significant pseudocapacitive behavior during discharge/charge processes, which partially accounts for the excellent lithium-ion storage performance.

Keywords Fe_2O_3@C composite · Iron concentrate leachate · ZIF-8 · Lithium-ion battery · Anode material

Introduction

With the continuous growth of global energy demand and increasingly prominent environmental issues, the development of efficient, clean and sustainable energy storage technologies has become a crucial direction in contemporary scientific research and industrial applications. Lithium-ion batteries (LIBs), owing to their high energy density, long cycle life and environmental friendliness, have dominated applications in electric vehicles, portable electronic devices, and smart grid systems [1–5]. However, the conventional graphite anode material exhibits a relatively low theoretical specific capacity (372 mAh g^{-1}), making it difficult to meet the requirements for high-energy-density storage [6]. Consequently, the development of novel high-performance anode materials has emerged as one of the current research priorities. Among various candidate anode materials, transition metal oxides (TMOs), particularly iron oxides (Fe_xO_y, including Fe_2O_3 and Fe_3O_4), have been recognized as one of the most promising anode materials for LIBs due to their superior theoretical capacity (~ 1000 mAh g^{-1}), abundant natural resources, low cost and environmental compatibility [7, 8]. Notably, iron oxide (Fe_2O_3) stands out for its distinctive merits. It delivers a theoretical specific capacity as high as 1007 mAh g^{-1}. Fe_2O_3 anode materials operate via a complete conversion reaction ($Fe_2O_3 + 6Li^+ + 6e^- \leftrightarrow 2Fe + 3Li_2O$), which significantly increases the theoretical specific capacity and reduces the initial lithiation potential to 0.8–1.0 V. However, iron oxide materials still suffer from several critical challenges in practical applications. For example, poor electrical conductivity, volume expansion and side reaction issues [9, 10]. To address these problems, researchers have proposed various modification strategies, including nanostructure design, carbon coating, conductive polymer compositing, and elemental doping [11, 12]. Among these methods, carbon coating has emerged as an

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effective methodology that simultaneously enhances electrical conductivity and mitigates volume expansion, thereby improving cycling stability. Metal-organic frameworks (MOFs) have been identified as ideal carbon precursors due to their high specific surface area, tunable pore structures and ability to form conductive carbon networks upon pyrolysis. Particularly, zeolitic imidazolate framework-8 (ZIF-8), which can be converted into porous carbon (PC) through thermal treatment, has been extensively employed in the fabrication of high-performance electrode materials owing to its exceptional electrical conductivity and chemical stability [13, 14]. Moreover, when employed as anode components for lithium-ion batteries, MOF-derived porous materials not only offer an elevated specific surface area and porosity to increase contact involving electrolytes and nanomaterials but also provide more active sites and shorter diffusion paths to accelerate lithium-ion diffusion.

Simultaneously, the efficient utilization of resources and sustainable development have emerged as global priorities. Iron concentrate, serving as the primary raw material for steel smelting, generates leachates rich in Fe^{2+}/Fe^{3+} ions during hydrometallurgical processing. Conventional treatment methods typically employ neutralization precipitation to recover iron residues, yet this approach demonstrates limited economic value-added potential and may potentially induce secondary pollution. The direct conversion of these leachates into high-value electrodes materials would not only facilitate resource recycling but also significantly reduce the production costs of electrode materials, thereby aligning with the fundamental principles of green chemistry and circular economy development [15, 16].

Hence, iron concentrate leachate was used as the iron source to prepare Fe_2O_3. Subsequently, the core-shell Fe_2O_3@C anode was prepared via a facile method with ZIF-8 as carbon sources. Benefitting from the hollow framework and carbon layer, the material illustrated excellent electrochemical performance. The initial discharge/charge capacity of the as-prepared Fe_2O_3@C electrode can reach 723.08 mAh g^{-1} at a current density of 1 A g^{-1}. The development of this method can provide a high-value-added conversion path for the waste liquid produced during the iron concentrate smelting process, which is expected to reduce the raw material cost of battery anode materials and decrease the generation of solid waste [17, 18].

Experimental section

Raw materials

All the reagents ($NH_3 \cdot H_2O$, $Co(NO_3)_2 \cdot 6H_2O$, PVP, $Zn(NO_3)_2$ and 2-methylimidazole) are all of analytically grade. The leachate of iron concentrate in this study was provided from a company situated in Yunnan, China. The main components of the leachate are shown in Table 1.

Preparation of Fe_2O_3 and Fe_2O_3@C

Fe_2O_3 was prepared at room temperature through a simple chemical precipitation method with iron concentrate leachate as the iron source and ammonia as the precipitant. A mixture of 0.5 g of Fe_2O_3 and 0.5 g of PVP was stirred in 100 mL methanol solution for 30 min. A certain mass of 2-methylimidazole was weighed and dissolved in methanol under ultrasonic treatment for 30 min until complete dissolution, which was denoted as Solution A. The pre-prepared Fe_2O_3, polyvinylpyrrolidone (PVP) and $Zn(NO_3)_2$ were added to 90 mL of methanol under continuous stirring until fully dissolved. Solution A was then introduced into the mixture and allowed to stand for 6 h to ensure complete coating of ZIF-8 onto Fe_2O_3. The product was washed with methanol, dried, and collected as the Fe_2O_3/ZIF-8 composite. The obtained composite was subsequently carbonized at 400 °C for 2 h under an argon atmosphere to yield the Fe_2O_3@C composite material. And the carbon content is only about 3%.

Material characterization

The phase structure of the as-prepared samples was characterized by an X-ray diffractometer (XRD, Austrian Anton Paar XRDynamic500). The morphology of the samples was examined using SEM (German ZEISS Fegma 300) and TEM (US FEI Talos F200x). X-ray photoelectron spectroscopy (XPS) was conducted using a Thermo Scientific K-Alpha analyzer to analyze the surface chemical composition and valence states. Nitrogen adsorption-desorption isotherms were measured using a US Micromeritics ASAP 2460 analyzer to determine the specific surface area of the samples.

Table 1 The main components of the leachate

Solutions	Constituents and content									
	Al	Co	Cr	Mn	Ti	K	Na	Ca	Mg	Fe
After leaching (g/L)	0.007	0.42	0.5	0.38	0.03	0.001	0.004	0.007	0.002	53.367

Electrochemical measurements

The electrochemical performance of Fe_2O_3 and Fe_2O_3@C were employed using CR2025 coin-type cells and tested at 25 °C within a voltage range of 0.01–3 V (vs. Li^+/Li) under varying current densities. CR2025 coin cells were assembled within an argon-filled glovebox. And 1 M $LiPF_6$ dissolved in a 1:1:1 volumetric mixture of ethylene carbonate (EC), dimethyl carbonate (DMC), and ethyl methyl carbonate (EMC) was used as the electrolyte. Fe_2O_3 or Fe_2O_3@C, PVDF, and Super P were mixed in a 7:2:1 mass ratio and dissolved in N-methyl-2-pyrrolidone (NMP) to form a uniform slurry. Subsequently, the slurry was applied to copper foil and vacuum-dried at 80 °C for 12 h. After drying, the foil was cut into circular electrodes with a 14 mm diameter. On the working electrode, the mass loading of active material was around 1.109 mg/cm². A polypropylene membrane served as the separator, while lithium foil was utilized as the counter electrode. And this electrode is used as the anode. Cyclic voltammetry (CV) analyses were performed using an electrochemical workstation (Chi 760 E, China) within a potential range of 0.01–3 V (vs. Li^+/Li) at scan rates of 0.1–1.1 mV s^{-1}. Electrochemical impedance spectroscopy (EIS) measurements were performed across a frequency range of 10 mHz to 100 kHz.

Result and discussion

The structure and composition of Fe_2O_3 and Fe_2O_3@C materials were carried out by XRD. As displayed in Fig. 1a. All synthesized samples exhibit no additional diffraction peaks compared to the reference pattern (PDF#33–0664), indicating the absence of impurity phases [19]. Furthermore, no detectable diffraction peaks corresponding to carbon were observed in sample Fe_2O_3@C, which is primarily due to the carbon content is too low to be detected. To directly examine the microstructural morphology of the material, scanning electron microscopy (SEM) analysis was conducted. As shown in Figs. 1b and c, Fe_2O_3 consists of rod-shaped particles with lengths ranging from 50 to 200 nm, which are randomly arranged and aggregated. However, Fe_2O_3@C is composed of numerous more uniform nanorods, with dimensions around 200 nm, as shown in Fig. 1d. Notably, a comparison between Figs. 1e and f reveals that Fe_2O_3@C exhibits larger interparticle voids. These voids can accommodate volume expansion during the lithiation and delithiation processes, thereby mitigating excessive electrode swelling and contributing significantly to the stability of electrochemical performance.

The TEM micrographs of Fe_2O_3 and Fe_2O_3@C are presented in Fig. 2. As can be observed from both Figs. 2a and c, materials Fe_2O_3 and Fe_2O_3@C consist of nanosheet-like particles, which is consistent with the results obtained from the SEM images. A comparison between Fig. 2b and d reveals that Fe_2O_3@C is coated with a uniform carbon layer, which is attributed to the high-temperature pyrolysis of ZIF-8. The thickness of this carbon layer is 2.36 nm, which significantly enhances the electrical conductivity of Fe_2O_3@C, thereby proving beneficial for its electrochemical performance. As shown in Figs. 2e-j, the distribution of Fe, O, N, Zn and C elements can be clearly observed, indicating that ZIF-8 is uniformly distributed on the surface of Fe_2O_3@C after high-temperature pyrolysis.

To investigate the specific surface area and pore structure characteristics of the composite material, we conducted N_2

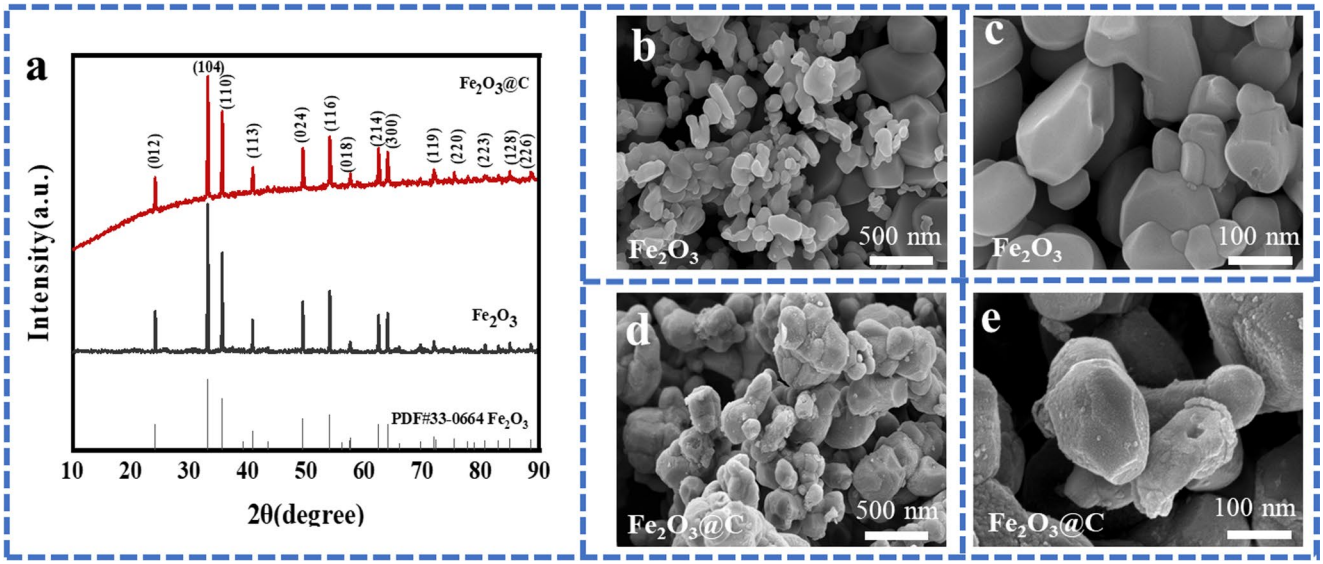

Fig. 1 XRD spectra (**a**) and SEM images of Fe_2O_3 and Fe_2O_3@C (**b-e**)

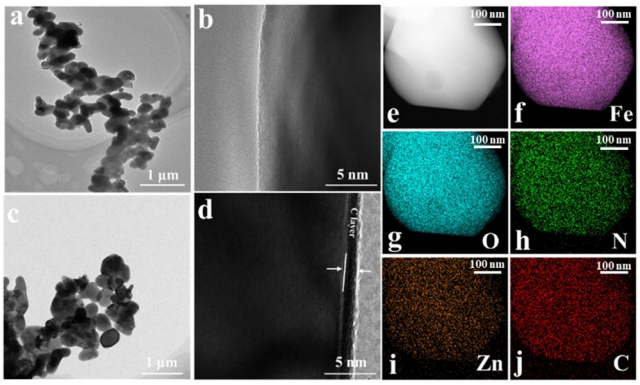

Fig. 2 TEM images of Fe_2O_3 at various magnifications (**a-c**); TEM images of Fe_2O_3@N-C at various magnifications (**d-f**); Elemental mapping images of Fe_2O_3@N-C (**g-j**)

adsorption tests, and the results are shown in Fig. 3. The specific surface area of the Fe_2O_3@C composite is determined to be 2.8153 $m^2\ g^{-1}$ and the pore size is 1.84 nm. Due to the porous nature of ZIF-8, after high-temperature carbonization, some pore structures appear. As a result, Fe_2O_3@C particles contain internal voids, which facilitates the infiltration of the electrolyte and alleviates the volume change of electrode material, thereby enhancing the electrochemical performance of the material [20, 21]. Figure 3c shows Raman test results for Fe_2O_3@C, D band at ~1180 and G band at ~1310 cm^{-1} correspond to disordered carbon and crystalline graphite respectively. Furthermore, the relatively high I_G/I_D ratio indicates a semi-graphitic carbon framework with excellent conductivity. Scattering peaks in the lower wavenumber range correspond to the nanostructure of Fe_2O_3.

The surface composition and bonding states of the Fe_2O_3@C material were determined by X-ray photoelectron spectroscopy (XPS). Figure 4a displays the XPS survey spectrum of Fe_2O_3@C, which confirms the presence of Fe, C, N, Zn and O elements in the sample. The high-resolution spectrum of Fe 2p is shown in Fig. 4b. The Fe 2p spectrum can be deconvoluted into two spin-orbit doublets corresponding to Fe $2p_{3/2}$ and Fe $2p_{1/2}$ levels. Main peaks at 711.0 eV and 724.1 eV are assigned to Fe^{3+} $2p_{3/2}$ and Fe^{3+} $2p_{1/2}$, respectively. Distinct satellite peaks are observed at higher binding energy sides of main peaks. The satellite at 718.6 eV is characteristic of Fe^{3+} in Fe_2O_3, arising from ligand-to-metal charge transfer. This satellite is key for distinguishing Fe^{3+} from Fe^{2+} or Fe^0 [22]. The high-resolution O 1 s spectrum (Fig. 4c) is deconvoluted into two component peaks. The peaks located at 530.1 and 531.1 eV are attributed to Fe–O and C–O bonds, respectively, indicating that Fe is covalently bonded to O. The high-resolution C 1 s spectrum (Fig. 4d) exhibits three peaks at 284.6 eV, 285.6 eV, and 288.3 eV. The peak at 284.6 eV is assigned to C–C bonds in the carbon layer coated on the surface of Fe_2O_3 nanoparticles. The peak at 285.6 eV is associated with C–O bonds. The peak at 287.8 eV is assigned to C–N bonds and the peak at 288.3 eV corresponds to C = O bonds. Four peaks are observed in the N 1 s spectrum, located at 410.06, 398.82, 397.31 and 394.17 (Fig. 4e), corresponding to oxidized nitrogen, pyrrolic nitrogen, pyridinic nitrogen and graphitic nitrogen, respectively. The successful doping of N into the C layer is indicated, leading to improved conductivity of the material [23, 24].

Figure 5a shows the cycling performance of materials Fe_2O_3 and Fe_2O_3@C over 300 charge-discharge cycles at a current density of 0.1 A g^{-1}. As can be seen from Fig. 5a, after 300 cycles, the reversible capacity of Fe_2O_3@C decreased from 1168.65 to 723.08 mAh g^{-1}, while that of Fe_2O_3 plummeted from 1204.12 to 205.37 mAh g^{-1}. The capacity retention rate of Fe_2O_3@C was 61.87%, whereas that of Fe_2O_3@C was only 17.06%. These results indicate that Fe_2O_3@C exhibits superior reversibility and cycling stability. The specific discharge capacity of the material shows an initial decrease followed by gradual increase, this increasing capacity upon cycling can be due to the refinement of electroactive materials, which could increase the electroactive area of active materials, offer additional interfacial charge storage, andinduce the unconventional activity

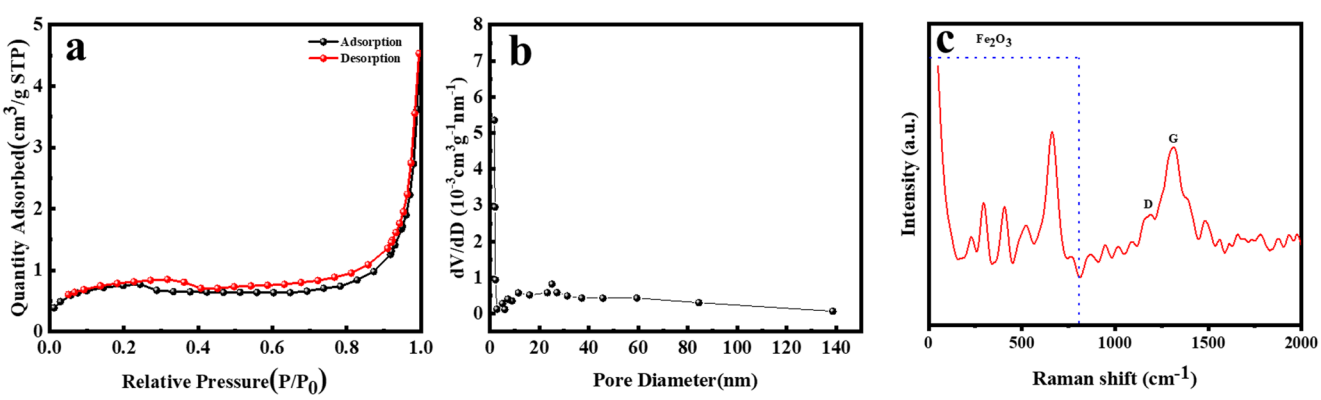

Fig. 3 The N_2 adsorption/desorption isotherms (**a**); Alongside the pore size distribution charts (**b**) for Fe_2O_3@C. Raman test (**c**)

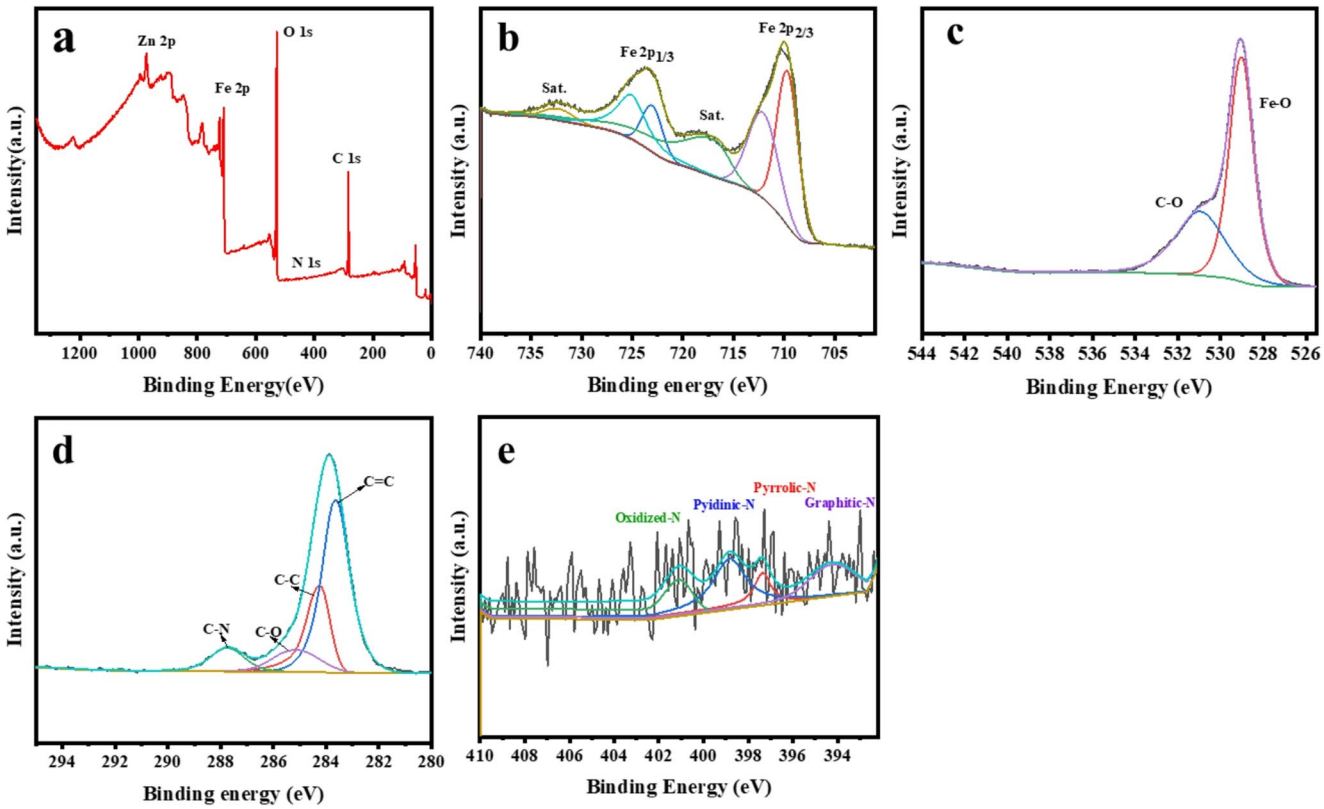

Fig. 4 (**a**) XPS survey spectrum and core-level XPS spectra of (**b**) Fe 2p (**c**) O 1 s and (**d**) C 1 s for the as-prepared Fe_2O_3@C sample

of structural defects. The rate performance of materials under varying current densities is presented in Fig. 5b. As clearly observed from the figure, Fe_2O_3@C demonstrates superior rate performance compared to Fe_2O_3. When the current density increases to 5 A g^{-1}, Fe_2O_3 exhibits poor reversible capacity, merely reaching 3.47 mAh g^{-1}. In contrast, under the same current density, Fe_2O_3@C maintains a capacity of 322.01 mAh g^{-1}. This enhancement can be attributed to the uniform coating derived from the high-temperature pyrolysis of ZIF-8 on the surface of Fe_2O_3@C, which results in a higher specific surface area, provides additional defects and porosity, and facilitates faster electron and ion transport during charge-discharge processes. The electrochemical performance comparison with previously reported Fe_2O_3 materials is presented in Table 2. Figures 5c and d present the distinct discharge/charge voltage profiles of electrodes Fe_2O_3 and Fe_2O_3@C under various current densities. As observed, these profiles exhibit similar variation trends, each comprising a rapid decline region (3–0.85 V), a plateau region (0.85 V), and a sloping region (0.75–0.01 V). In Fig. 5c, as the cycle number increases from 1 st to 300th, noticeable attenuation occurs in both the plateau and sloping regions. This indicates that electrode Fe_2O_3 fails to adequately sustain the conversion reaction and surface interfacial processes at elevated current densities. In contrast, electrode Fe_2O_3@C maintains a high and stable capacity, with nearly identical plateau regions across different cycle number and a reversible capacity in the sloping region still exceeding 600 mAh g^{-1}, demonstrating its superior interfacial lithium-ion storage capability.

To further investigate the kinetic differences between Fe_2O_3 and Fe_2O_3@C during the charging and discharging processes, electrochemical impedance spectroscopy (EIS) was conducted over a frequency range of 0.1 Hz to 100 kHz. The results are presented in Fig. 6. The Nyquist plots of both electrodes exhibit a similar pattern, consisting of a depressed semicircle in the medium-to-high frequency region and a sloped line in the low-frequency region. The semicircle corresponds to the charge transfer resistance and contact impedance, while the sloped line is associated with ion diffusion between the electrolyte and the electrode surface [30, 31]. Notably, the diameter of the depressed semicircle for Fe_2O_3@C is significantly smaller than that of Fe_2O_3, indicating lower contact and charge transfer resistances. These results suggest that the carbon layer formed on the surface of Fe_2O_3 after the high-temperature pyrolysis of introduced ZIF-8 can enhance the electrical conductivity and electrochemical activity of the electrode material, which is crucial for electron transfer and performance during lithium-ion cycling. Table 3 summarizes the EIS fitting parameters, Warburg factors, and ionic conductivity values for Fe_2O_3 and Fe_2O_3@C.

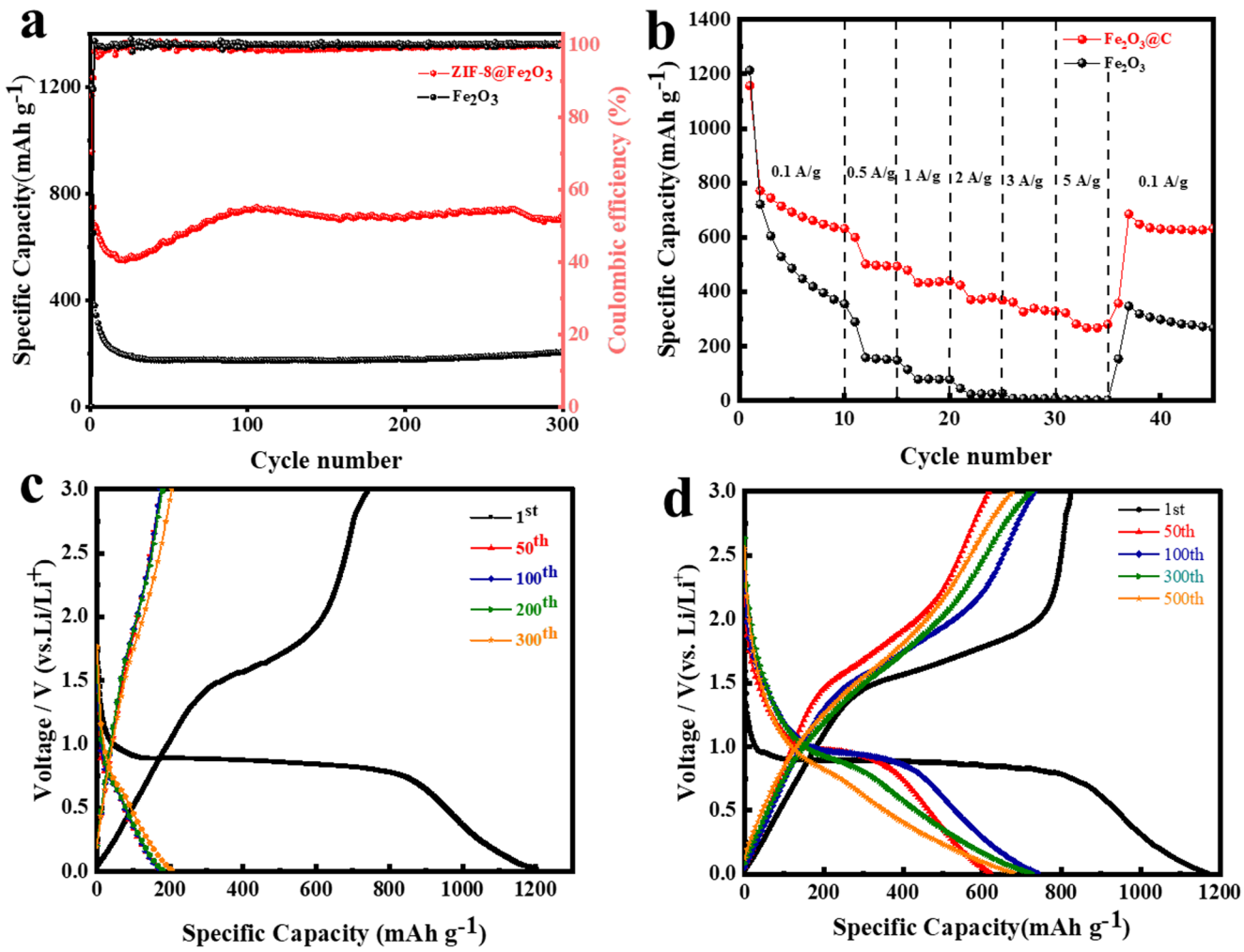

Fig. 5 Cycle stabilities (**a**) and rate capabilities (**b**) of Fe_2O_3 and Fe_2O_3@C at 0.1 A/g; Galvanostatic charge/discharge profiles of Fe_2O_3 (**c**) and Fe_2O_3@C (**d**) in the different cycles at a current density of 0.1 A g^{-1}

Table 2 Electrochemical performance comparison with references

Sample	Capacity(mAh g^{-1})	Ref
SnO_2/Fe_2O_3	598.08 mA h g^{-1} over 100 cycles at 0.4 A g^{-1}	[25]
Si/Fe2O3	697.2 mA h g^{-1} over 100 cycles at 0.2 A g^{-1}	[26]
Si@Fe_2O_3/C	680.7 mA h g^{-1} over 300 cycles at 1 A g^{-1}	[27]
Fe_2O_3/CNTs	619 mA h g^{-1} over 80 cycles at 0.05 A g^{-1}	[28]
Fe_2O_3/OMWCNTs	653.6 mA h g^{-1} over 200 cycles at 0.5 A g^{-1}	[29]
Fe_2O_3@C	723.08 mA h g^{-1} over 300 cycles at 1 A g^{-1}	This worke

We investigated the electrochemical kinetics of Fe_2O_3 and Fe_2O_3@C by employing cyclic voltammetry (CV) at different scan rates. The results are presented in Figs. 7a and e. It can be observed that both the oxidation and reduction peaks progressively shift as the scan rate increases. To qualitatively analyze this behavior, the relationship between the current and the scan rate was derived from the CV curves obtained at various scan rates, as described by Eqs. (1) and (2) below [32].

$$i = av^b \tag{1}$$

$$log(i) = b\ log(v) + log(a) \tag{2}$$

Where, i and v represent the peak current and scan rate, respectively, and a is a constant. The value of b was determined by plotting the fitted line of log(i) versus log(v) and calculating its slope. If b value close to 0.5 suggests a process that is primarily diffusion-controlled, while a value approaching 1.0 indicates capacitive behavior [33].

As illustrated in Figs. 7b and d, the b-values were determined through curve fitting. The calculated b-values for Fe_2O_3 and Fe_2O_3@C are 0.454 and 0.7866, respectively. These results indicate that the Li^+ storage process in Fe_2O_3

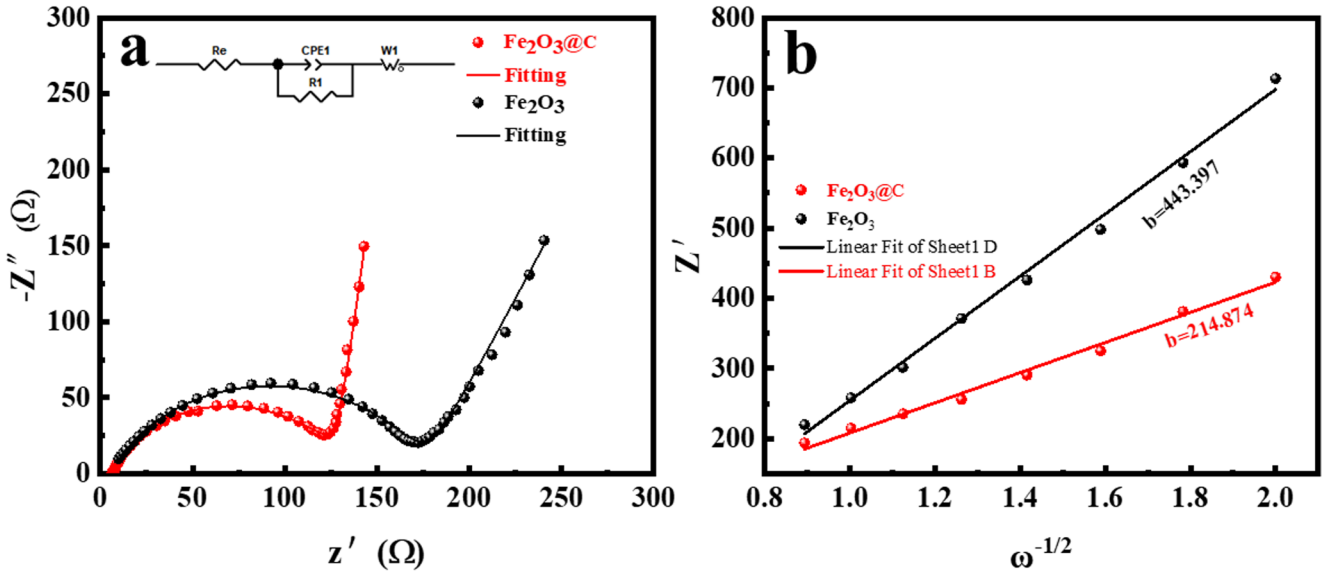

Fig. 6 The Nyquist diagrams (**a**) and the straight-line correlation between Z' and $\omega^{-1/2}$ (**b**) for Fe_2O_3 and Fe_2O_3@C

Table 3 Summary of the EIS fitted parameters, ion conductivity and Warburg factor of the samples

Samples	Fitted parameters			
	R_e/(Ω)	R_1/(Ω)	σ/(Ω $s^{-1/2}$)	D/(cm^2s^{-1})
Fe_2O_3	5.26	123.8	468.397	6.48×10^{-12}
Fe_2O_3@C	4.24	47.02	50.286	7.21×10^{-11}

is primarily controlled by diffusion behavior, whereas in Fe_2O_3@C it is dominated by capacitive behavior. The specific percentage of the pseudocapacitive contribution can be quantified using Eq. (3) [34, 35].

$$i(V) = k_1 v + k_2 v^{1/2} \tag{3}$$

To elucidate the specific contributions of the electrodes at fixed scan rates, the respective contributions of pseudocapacitive (k_1v) and diffusion-controlled $\left(k_2 v^{\frac{1}{2}}\right)$ processes to the total current at a fixed potential (v) were deconvoluted. The values of k_1 and k_2 can be readily determined by fitting the plot of $\frac{i(v)}{v^{\frac{1}{2}}}$ versus $v^{\frac{1}{2}}$ [36–38]. As observed in the CV curves presented in Figs. 7c and g, at a scan rate of 1 mV s^{-1}, approximately 60.10% and 89.17% of the total capacity for Fe_2O_3 and Fe_2O_3@C, respectively, are attributed to capacitive contributions. Furthermore, the proportion of capacitive contribution increases correspondingly with increasing scan rate (Figs. 7d and h). The bar charts clearly illustrate that for Fe_2O_3, the pseudocapacitive contribution rates at scan rates of 0.2, 0.4, 0.6, 0.8, and 1.0 mV s^{-1} are 22.87%, 31.16%, 38.59%, 46.48%and 60.10%, respectively, while those for material Fe_2O_3@C is 65.17%, 72.48%, 81.18%, 85.86%and 89.17%. These results indicate that at high current densities, the capacity of the electrode materials is predominantly governed by surface-driven pseudocapacitive behavior.

Conclusion

In conclusion, we have successfully synthesized the Fe_2O_3@C nanoparticles through a simple chemical precipitation process, employing iron concentrate leachate as the source of iron. The use of ZIF-8 of Fe_2O_3@C sample improved the electrical conductivity and lithium-ion diffusion kinetics by providing porous structures, resulting in improved electrochemical performance. The reversible capacity of Fe_2O_3@C decreased from 1168.65 to 723.08 mAh g^{-1} after 300 cycles. The capacity still exhibited as high as 322.01 mA h g^{-1} even at a high rate of 5 A g^{-1}. This study confirms that ZIF-8-derived carbon layers achieve comprehensive enhancement through four synergistic mechanisms: (1) Conductive network: Forms continuous conductive matrix, significantly reducing charge transfer resistance, ensuring excellent rate performance. (2) Conductive network: Forms continuous conductive matrix, significantly reducing charge transfer resistance, ensuring excellent rate performance. (3) Interface optimization: Porous structure and N-doping synergistically enhance ion transport and induce pseudocapacitance, improving capacity and fast-charging capability. (4) Interface stabilization: Uniform carbon layer acts as physical barrier, stabilizes SEI, reduces side reactions, improves Coulombic efficiency and long-term cycling life. This work can provide useful guidance for the development of preparation of electrode materials from solid waste.

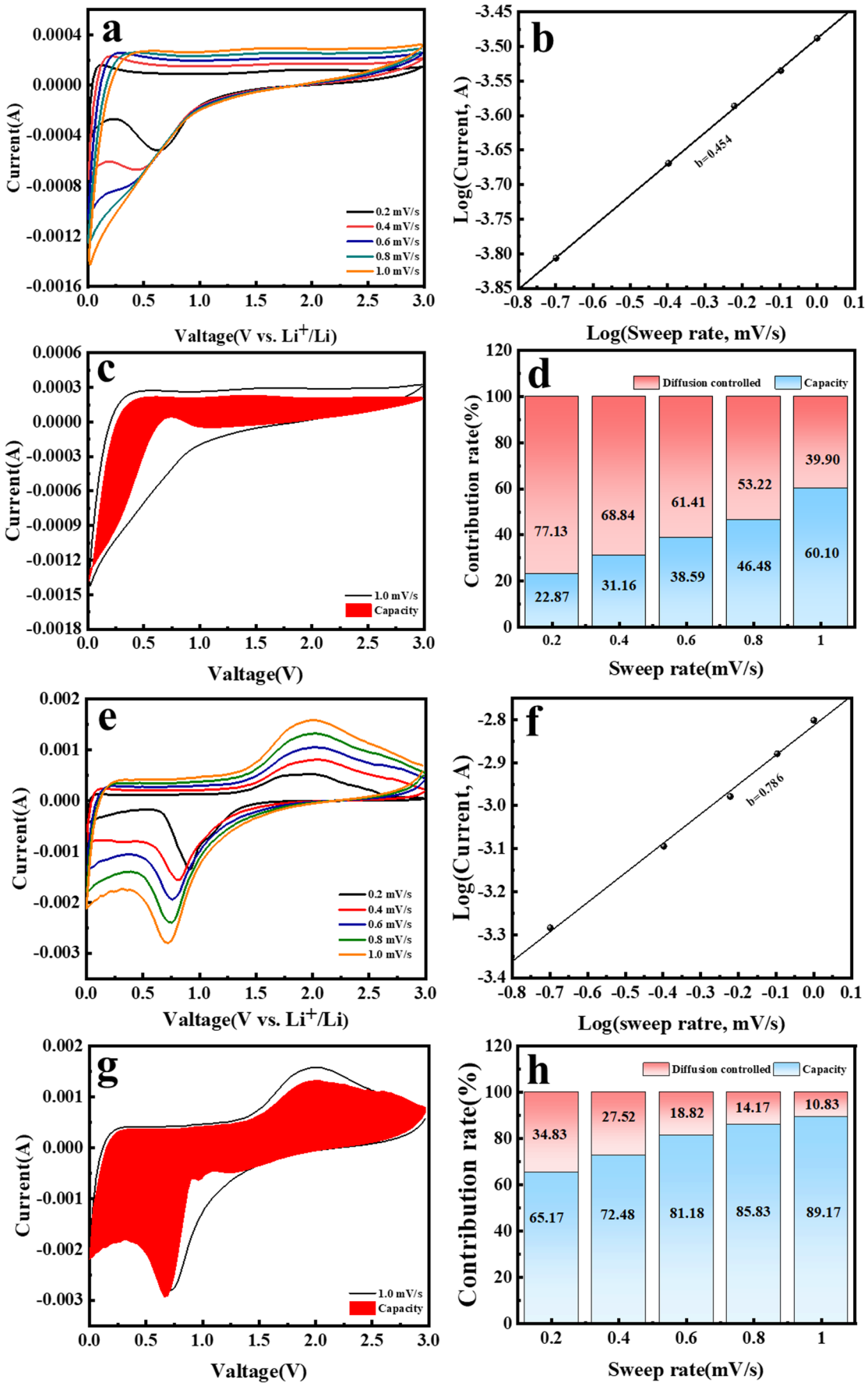
a
Current(A)
Valtage(V vs. Li+/Li)
0.2 mV/s
0.4 mV/s
0.6 mV/s
0.8 mV/s
1.0 mV/s
b
Log(Current, A)
Log(Sweep rate, mV/s)
b=0.454
c
Current(A)
Valtage(V)
1.0 mV/s
Capacity
d
Contribution rate(%)
Sweep rate(mV/s)
Diffusion controlled
Capacity
77.13
22.87
68.84
31.16
61.41
38.59
53.22
46.48
39.90
60.10
e
Current(A)
Valtage(V vs. Li+/Li)
0.2 mV/s
0.4 mV/s
0.6 mV/s
0.8 mV/s
1.0 mV/s
f
Log(Current, A)
Log(sweep ratre, mV/s)
b=0.786
g
Current(A)
Valtage(V)
1.0 mV/s
Capacity
h
Contribution rate(%)
Sweep rate(mV/s)
Diffusion controlled
Capacity
34.83
65.17
27.52
72.48
18.82
81.18
14.17
85.83
10.83
89.17

Fig. 7 CV curves of the Fe_2O_3 sample at various scan rates (**a**); Linear relationship between the log (peak current) and log(scan rate) of Fe_2O_3 sample (**b**); The percentage contribution of the pseudo-capacitor pseudocapacitive (1.0 mV s^{-1}) (**c**); contribution ratio to the total capacity of the Fe_2O_3 sample at different scan rates (**d**). CV curves of the Fe_2O_3@C sample at various scan rates (**a**); Linear relationship between the log (peak current) and log(scan rate) of Fe_2O_3@C sample (**b**); The percentage contribution of the pseudo-capacitor pseudocapacitive (1.0 mV s^{-1}) (**c**); contribution ratio to the total capacity of the Fe_2O_3@C sample at different scan rates (**d**)

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Authors' contributions Guimin Zhou and Yin Li worte the main manuscript text. Li Wang, Yaochun Yao and Shaoze Zhang prepared Figs. 1, 2 and 3. Keyu Zhang, Junxian Hu, Zhunqin Dong and Lingling Yuan prepared Figs. 4, 5, 6 and 7. All authors reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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